FLEET TEAM DIGEST

737MAX-FTD-32-22001

Issue Title : Rotor Clip and Rivet Damage on Safran Brake Assembly

Airplane Model	ATA	Minor Model(s)
737MAX	3241-41	-8, -8200, -9, -BBJ

Other Model(s)	ECCN
	9E991

Originated Date	Last Revised Date	Created On
02/07/2022	07/29/2025	07/29/2025

Estimated Completion Date	Next Update Date

Revision Description

Minor text improvements.

Applicability

737 MAX airplanes using Safran Landing System (SLS) Carbon Brake Assy P/N C20633000.

Description

Boeing has received reports from 737NG and MAX Operators of finding partial or complete cracks on some/multiple rotor clips on Brake Assembly P/N C20633000. This phenomenon happens mainly on 737MAX fleet, the 737NG can also experience this as the brake assembly reaches near its end of life as indicated by the wear indicator pin. These cracks are in general on the edge of the clip (vertical) and usually on the first rotor of the carbon brake. In rare cases, some cracks are visible on the second rotor, missing rivets on the rotor clips, or damaged rivets have also been observed as shown in Figure 1 and 2 attached within this Fleet Team Digest (FTD) article.

Status

Safran Landing Systems (SLS) has determined that installation of radial tires causes the wheel to deform in such a way that the wheel torque bars make uneven contact with the brake rotor clips. This uneven contact is causing fatigue damage to the hardware causing the clip and/or attaching hardware to crack.

Note: This uneven contact can occur on the 737NG fleet with bias tires as well, however, if this does occur, it usually manifests itself when the wear pin indicator for the brake assembly is near its end of life.

In early 2022, SLS developed an improved rotor clip design to prevent rotor clip cracking and rivet damage. The improved rotor clip design failed qualification testing during its last month of testing in November 2022. SLS conducted a failure analysis to identify the root cause of the qualification test failure and make redesign adjustment to the rotor clip design. Due to the first improvement design failure, Boeing has asked SLS to study two potential designs (version 2 option ‘A’ & version 2 option ‘B’) to

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mitigate any additional delays should one design not fulfill all of the qualification requirements and/or have superior performance differences. SLS has determined that version 2 option ‘B’ has the ideal performance which will be used to finalize all testing to mitigate the rotor clips from cracking.

Interim Action

Please note, when the wheel-tire assembly will be installed on the landing gear axle per AMM Task 32-45-11-400-801, it is required to perform an inspection on the brake assembly per paragraph G.(1).(d). Boeing emphasizes the importance of thoroughly inspecting the brake assembly during the removal of the wheel-tire assembly change per 737 MAX AMM Pageblock 32-45-11/401, 'Main Landing Gear Wheel And Tire Assembly - Removal/Installation'.

When performing AMM Task 32-41-41-700-803 'Main Landing Gear Brake Inspection (Wheel Removed from the Airplane)', ensure to examine the brake for these types of damage:

- Cracks that you can see, this includes inspecting each rotor for cracked clips
- Broken parts, this includes inspecting each rotor clip for broken rivets and rivet heads
- Do a check for loose rotor drive clips and rivets
- Examine the brake and surrounding area for unusual carbon debris such as chips or small pieces of carbon broken off of the rotors and stators

If damage is found as shown in the attached file within this FTD article, Boeing recommends replacing the brake assembly. If the brake assembly cannot be replaced, Boeing can provide a small number of flight cycles in order to reach the maintenance facility to remove the brake assembly. For such request, please submit a Service Request (SR) to Boeing containing clear and detailed photographs of the brake assembly for review.

Final Action

SLS has completed its redesign of brake assembly P/N C20633000. The redesign consists of replacing the H-rotor clip with a U-rotor clip, as well as resizing the attaching rivet and removing the scallop from the carbon rotor lug. This brake redesign will be incorporated into production starting at LN 9259.

For in-service airplanes before LN 9259, SLS has created service bulletin C20633-32-025 which replaces the existing brake assembly rotors (contains H-clips) with the enhanced rotors (contains U-clips). Once the service bulletin has been incorporated, the existing brake assembly Amdt letter A or B or C shall be classified as Amdt letter D for brake assembly P/N C20633000. Boeing recommends incorporating the SLS service bulletin during a convenient maintenance opportunity in order to mitigate future rotor clip cracks or missing rivets on the brake assembly. For more information regarding service bulletin C20633-32-025, please contact Safran Landing Systems.

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Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

- Solution Selected: September 2024 (Completed)
- Qualification Testing & Final Report: April 2025 (Completed)
- Boeing production line: July 2025 (Completed)
- Solution available to the In-service fleet: July 2025 (Completed)

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Aircraft Maintenance Manual (AMM)	32-45-11		
Aircraft Maintenance Manual (AMM)	32-41-41		
Illustrated Parts Catalog (IPC)	32-41-41-01		
Illustrated Parts Catalog (IPC)	32-41-41-11		
Component Maintenance Manual (CMM)	CMM 32-49-80		
Fleet Team Digest Article (FTD)	737NG-FTD-32-22002		
Service Bulletin (SB)	C20633-32-025		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Brake Assy	C20633000	Safran Landing Systems	VF6137

Related Categories

- AOGs : Events that cause a significant Airplane On Ground (AOG) condition
- Best Practices / Operational Efficiency : Operational / Maintenance-focused best practices, techniques, health checks or recommendations for preserving aircraft efficiency.
- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

Attachments:

a) Example Photographs

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